

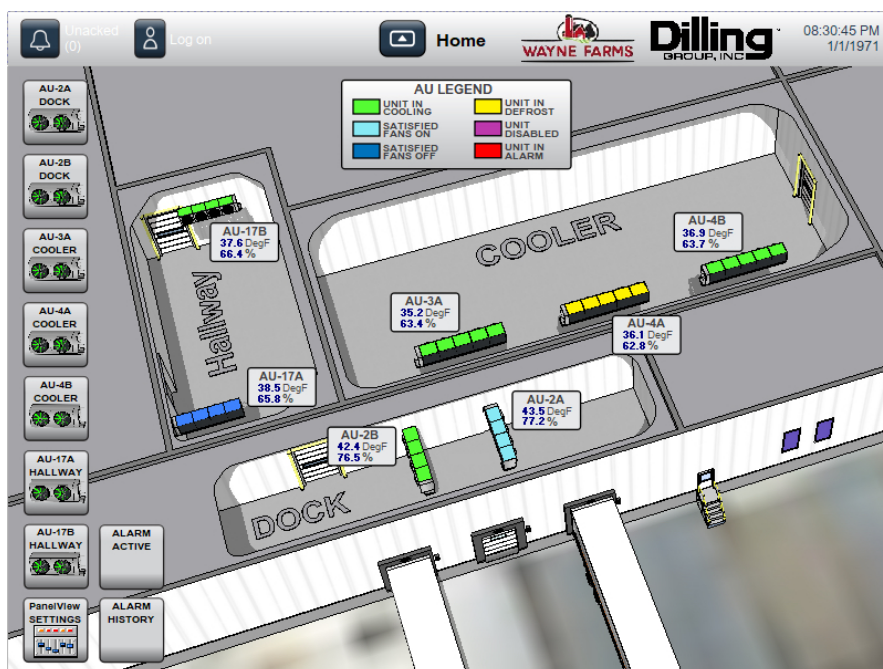
REFRIGERATION CONTROLS AIR UNIT ADDITION 2021 AU-2A, 2B, 3A, 4A, 4B, 17A & 17B MANUAL

REFRIGERATION SYSTEM CONTROLLER

The refrigeration system control is based on Allen-Bradley's Compact Logix's 5380 platform. This controller is connected to all of the field equipment Input/Output (I/O) points for controlling and monitoring of equipment. The operator interface located on the panel door an Allen-Bradley Panelview 5310, 10" color touch screen. The PanelView is connected to the PLC via Ethernet and can be connected to the plant network for remote access.

MAIN "HOME" SCREEN AND SYSTEM BUTTONS

The Main Screen is the base screen for accessing and viewing the entire system. Critical systems information (i.e. area temperatures, humidity and status) are displayed on this screen. Differing modes and status for equipment are represented by various colors. Below is an example of the "Main Screen" and with a Legend that shows the meaning of the colors for Air Units.



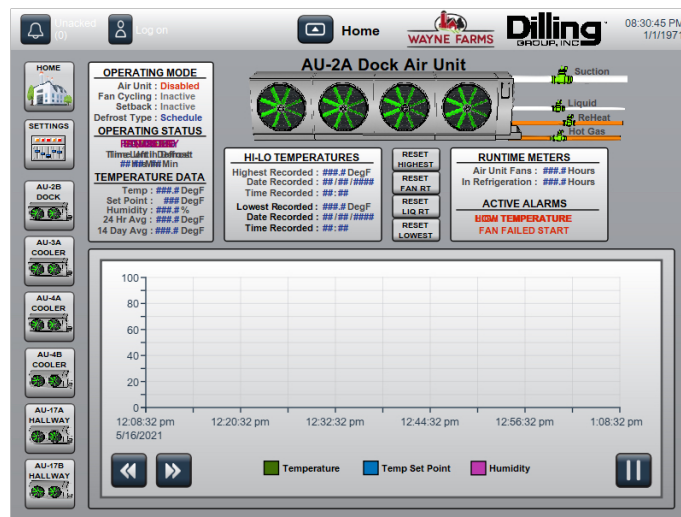
Main Screen

To view an Air Unit in greater detail select the desired button located down the left side of the screen. All full screens have the Banner located along the top. This banner allow an operator to return to the Main screen by selecting "Home". Log into the system by selecting "Log On" default user name is WF, password is 1000. Users can be added during startup. Also acknowledge Alarms as well as see if there are any active alarms on the first button.

Evaporative Air Units Control

Detail Screen

From the main screen each air unit Detail Screens can be selected by pressing the desired air unit button on the left side of the screen. From this screen you can see detailed information about an air unit including, 1-hour trend, Valve status, Fan status, Runtime meters for fans and for unit cooling. Operating Mode information, Temperature Data including 24 hour & 14 day running average, Time until next defrost, High/Low Temperature, time and date of highest and lowest values recorded. Runtime hours and High/Low temperature data can be reset once logged in.



Evaporator Detail Screen

Automatic Control

Each evaporator is controlled from a temperature/humidity sensor located in the air return side of each unit. A temperature set point for each unit is entered at the Graphics Interface Computer. The system controller will add one (1) and subtract one (1) from the set point that is entered to give a two (2) degree dead band. The reheat solenoid operates from a humidity setpoint and will remain on until the humidity has decreased by 2%. If the room humidity becomes greater than setpoint the reheat will come on "energize" until it reaches the offset. There are four (4) solenoid valves for each evaporator, individually controlled depending on the mode of the evaporators as described below. The units fan starters are controlled together with one (1) output from the controller at the Engine Room MCC.

A Satisfied temperature condition is true when the return air temperature for an evaporator is below the two (2) degree dead band. The units will remain Satisfied until the temperature has risen above the two (2) degree dead band.

Controller Action for Satisfied Units:

- Fans On or may cycle if Fan Cycling is enabled
Fan Cycling should be disabled for Humidity Control.
- Liquid Valve Closed (de-energized).
- Suction Valve Open (energized).
- Hot Gas Valve Closed (de-energized).
- Re-Heat Valve, controlled by Humidity setpoint.

A Call for Cooling condition is true when the return air temperature is above the two (2) degree dead band. The units will continue to cool until the temperature falls below the two (2) degree dead band. The two-degree dead band is plus and minus one from the setpoint.

Controller Action for Units Cooling:

- Fans On, will override Fan Cycling.
- Liquid Valve Open (energized).
- Suction Valve Open (energized).
- Hot Gas Valve Closed (de-energized).
- Re-Heat Valve, controlled by Humidity setpoint.

Defrost Control

The Evaporators have Hot Gas to defrost the coils. The following is the defrost sequence and controller action which each of the units will go through when defrost is initiated:

Pump Out Cycle:

Pumpout will remove any existing ammonia from the coil and is typically set from 5 minutes to 30 minutes

- Fans On
- Liquid Valve Closed (de-energized).
- Suction Valve Open (energized).
- Hot Gas Valve Closed (de-energized)
- Re-Heat Valve Open (energized) If unit is calling for Re-Heat.

Hot Gas Cycle:

Removes any frost/ice that has formed on the coil and is typically set from 10 minutes to 30 minutes.

- Fans Off
- Liquid Valve Closed (de-energized).
- Suction Valve Closed (de-energized).
- Hot Gas Valve Open (energized)
- Re-Heat Valve Open (energized) If unit is calling for Re-Heat.

Equalize/Bleed Cycle:

Brings the coil back down from Discharge (hot gas) pressure to suction pressure and is typically set from 3 minutes to 10 minutes.

- Fans Off
- HS Liquid Valve Closed (de-energized).
- HS Suction Valve Closed (de-energized).
- Hot Gas Valve Closed (de-energized)
- Re-Heat Valve Open (energized) If unit is calling for Re-Heat.

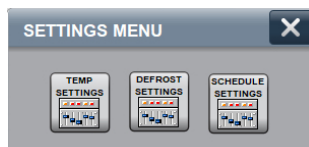
Fan Delay Cycle Mode:

Chills the coil back down and freezes any water droplets on the coil before turning the fans back on. Typically set from 3 minutes to 5 minutes.

- Fans Off
- Liquid Valve Open (energized).
- Suction Valve Open (energized).
- Hot Gas Valve Closed (de-energized)
- Re-Heat Valve Open (energized) If unit is calling for Re-Heat.

Settings

All Air unit settings can be changed by selecting the “Settings” Button located on the left side of each air units Detail Screen. The operator must be logged in to access these screen, the settings are broke down into the following three popup screens once “settings” is selected.



Settings Menu Screen

Temperature Settings

When the "Temp Settings" button is selected the following screen will popup.

The screenshot shows a software interface titled "AU-2A TEMP & MISC SETTINGS". It contains the following sections:

- Temp/Humidity Settings:** Includes input fields for "Temp Setpnt: ####", "Hi Temp Alarm: ####", "Lo Temp Alarm: ####", and "Humidity Setpnt: ####".
- Unit Enable/Disable:** Features a red "DISABLED" button.
- Humidity Override:** Includes a "Temp Offset: ####" field (with a note "Temp Above Set Point") and a red "DISABLED" button.
- Fan Cycling:** Includes input fields for "Fan ON Time: ####" and "Fan OFF Time: ####", a green "DISABLED" button, and a note "Enter 0 in ON time to keep fans off".

Temperature and Misc Settings Screen

Temp/Humidity Setting Values

These set points are used for operation and alarm settings.

The Temp Set Point value can be set from -40 to 140 Degree's F

The Hi Temp Alarm value can be set from -40 to 140 Degree's F

The Lo Temp Alarm value can be set from -40 to 140 Degree's F

The Humidity value can be set from 0 to 100 %

The Humidity Override value can be set from 3 to 7 Degree's F

Temperature Set Points

The system controller will add one (1) and subtract one (1) from the set point that is entered to give a two (2) degree dead band. When the temperature is above this deadband the controller will put the unit in cooling and will remain in cooling mode until the temperature gets 1 degree below the set point.

Humidity Set Points

The system controller will add one (1) and subtract one (1) from the set point that is entered to give a two (2) % dead band. When the humidity is above this deadband the controller will energize the Re-Heat coil and re-heat will remain on until the humidity gets 1 % below the set point.

Humidity Override can be enabled or disabled with the button shown above. If enabled it will stop the unit Re-Heat from being energized "On" when the area temperature is "Temp Offset" value above the Temp set point. The Reheat will not be allowed back on until the temperature reaches set point.

Unit Enable/Disable

When an Air Unit is Disabled it will go through a 5-minute Pumpout then all solenoids are De-Energized and fans turned off until its Enabled again. The toggle switches in control panels will override this function

Fan Cycling Settings

If Fan Cycling is not enabled the fans will remain on at all times (except defrost). If Fan Cycling is enabled and the Evaporator is satisfied the fans will run for the amount of time entered into the On time and will cycle off for the amount of time entered into the Off time. Upon a call for cooling the fans will be turned on even if currently off from Fan Cycling. To keep fans off when an air unit is satisfied a zero must be entered in the fan cycle on time.

The Fan On Time Value can be from 0 to 120 Minutes

The Fan Off Time Value can be from 1 to 120 Minutes

Defrost Settings

When the "Defrost Settings" button is selected the following screen will popup. The operator can enter the following values for the defrost cycle under Enter Values in Minutes.

The Pump Out Value can be set from 1 to 20 Minutes

The Hot Gas Value can be set from 1 to 60 Minutes

The Bleed Value can be set from 1 to 20 Minutes

The Fan Delay Value can be set from 1 to 20 Minutes

Demand Defrost

The operator can select whether to use Demand or Scheduled Defrost. By selecting the button below Defrost Type the system will toggle between these two modes. If the Demand Defrost is enabled the controller will accumulate the amount of time which has elapsed in the cooling cycle (Liquid Valve On Time). When this accumulated time reaches the Demand Interval Set Point, the unit will begin the defrost cycle. The Demand Defrost accumulated time will be reset upon finishing the defrost cycle.

The Demand Defrost interval Value can be set from 1 to 168 Hours

The screen displays the 'AU-2A DEFROST SETTINGS' window. At the top, there are four input fields for 'Enter Values in Minutes': Pump Out, Hot Gas, Equalize, and Fan Delay, each with a '####' placeholder. To the right is a 'Force Defrost' section with a 'START DEFROST' button. Below these is the 'Defrost Type' section, which has a 'DEMAND' button selected. Underneath is an 'Enter Value in Hours' field with a '####' placeholder.

Demand Defrost Settings

Scheduled Defrost

The Scheduled Defrost function allows a seven day defrost schedule to be entered at the operator interface. In this mode an evaporator can be scheduled to defrost up to four (4) times a day. The Defrost Type button is used to choose between a scheduled defrost or a demand type defrost. Schedules for each day of the week must be enabled and time to defrost entered in Military Time. If all defrost schedules are left disabled the unit will not defrost.

The screen displays the 'AU-2A DEFROST SETTINGS' window. The 'Defrost Type' section now has a 'SCHEDULE' button selected. Below this is a grid for scheduling defrost events. The grid has columns for days of the week (Mon, Tues, Wends, Thurs, Fri, Sat, Sun) and rows for four schedules (Sch1, Sch2, Sch3, Sch4). Each cell in the grid contains a '####' placeholder and a 'DISABLED' label.

Scheduled Defrost Settings

Force Defrost

The Force Defrost function will allow for an Evaporator to be put into Defrost at any time.

Note: Only one evaporator can go into defrost in at a time so if another is defrosting it will wait until other units are finished.

Defrost Groups

The seven air units controlled from this panel are tied together in one desrost group. Only one of these units can be in defrost at a time.

Temperature Setback

When the "Schedule Settings" button is selected the following screen will popup The Temperature Setback function will enable the Evaporators in each area to be set to a different temperature control point at a specific time for a specified duration. There are seven Setback Schedules available to use and must be enabled "below each schedule" in order for that schedule to be active. Below is a description for Schedule 1's settings.

Temp: Enter the new temperature setpoint desired during the scheduled period.

Day In: Select what day of week you want the schedule to start.

Time In: Enter what time on Day In you want the schedule to start, Military time 0 to 2359

Day Out: Select what day of week you want the schedule to stop.

Time In: Enter what time on Day Out you want the schedule to stop, Military time 0 to 2359

Enable/Disable: button for Schedule 1

Setback "Schedule Settings"

Time Delay After Power Failure

Upon a power failure the Evaporators will delay sixty (60) seconds between each evaporator before restarting when power has been returned to the panel.

Manual Control

A Selector Switch "Manual Power Feeds" on the control panel door enables/disables power to the manual side of the Switch Banks located in the Air Unit Control Panel. This Switch should only be in the "ON" position if air units are being controlled manually. The Toggle HOA switch banks enable manual control of each air units valves and fans. These switches must be placed in appropriate position for the desired refrigeration cycle (cooling/defrost). This feature should only be used if the PLC fails or some other fault condition causes the operator to need to override all automatic controls. Normal operation all Switches should be in the Auto position.

Manual Cooling

Switch Bank Positions for Unit Cooling:

- Liquid Valve Switch Bank toggle in HAND (energized).
- Suction Valve Switch Bank toggle in HAND (energized).
- Hot Gas Valve Switch Bank toggle in OFF (de-energized).
- Re-Heat Valve Switch Bank toggle in HAND or OFF.
- Fans Switch Bank toggle in HAND (fans running).

Switch Bank Positions for Unit Satisfied:

- Liquid Valve Switch Bank toggle in OFF (de-energized).
- Suction Valve Switch Bank toggle in HAND or OFF.
- Hot Gas Valve Switch Bank toggle in OFF (de-energized).
- Re-Heat Valve Switch Bank toggle in HAND or OFF.
- Fans Switch Bank toggle in HAND or OFF.

Manual Defrost

Pump Out Cycle:

Pumpout will remove any existing ammonia from the coil and is typically set from 5 minutes to 30 minutes

- Liquid Valve Switch Bank toggle in OFF (de-energized).
- Suction Valve Switch Bank toggle in HAND (energized).
- Hot Gas Valve Switch Bank toggle in OFF (de-energized).
- Re-Heat Valve Switch Bank toggle in OFF (de-energized).
- Fans Switch Bank toggle in HAND (fans running).

Hot Gas Cycle:

Removes any frost/ice that has formed on the coil and is typically set from 10 minutes to 30 minutes.

- Liquid Valve Switch Bank toggle in OFF (de-energized).
- Suction Valve Switch Bank toggle in OFF (de-energized).
- Hot Gas Valve Switch Bank toggle in HAND (energized).
- Re-Heat Valve Switch Bank toggle in OFF (de-energized).
- Fans Switch Bank toggle in OFF (fans off).

Equalize/Bleed Cycle:

Brings the coil back down from Discharge (hot gas) pressure to suction pressure and is typically set from 3 minutes to 10 minutes.

- Liquid Valve Switch Bank toggle in OFF (de-energized).
- Suction Valve Switch Bank toggle in OFF (de-energized).
- Hot Gas Valve Switch Bank toggle in OFF (de-energized).
- Re-Heat Valve Switch Bank toggle in OFF (de-energized).
- Fans Switch Bank toggle in OFF (fans off).

Fan Delay Cycle Mode:

Chills the coil back down and freezes any water droplets on the coil before turning the fans back on. Typically set from 3 minutes to 5 minutes.

- Liquid Valve Switch Bank toggle in ON (energized).
- Suction Valve Switch Bank toggle in ON (energized).
- Hot Gas Valve Switch Bank toggle in OFF (de-energized).
- Re-Heat Valve Switch Bank toggle in OFF (de-energized).
- Fans Switch Bank toggle in OFF (fans off).

Alarms\Warning

Operator adjustable High Temperature Alarm set point.

Operator adjustable Low Temperature Alarm set point.

Fan Fail Start Alarm, fan told to start and no auxiliary conformation back from the starter.

Temperature Sensor Failed Over/Under Range Alarm.

Humidity Sensor Failed Over/Under Range Alarm.

Unit Disabled Warning.

Parts List Air Unit Control Panel

See prints for Item No. locations

<i>Item No.</i>	<i>Qty.</i>	<i>Description</i>	<i>Part Number</i>	<i>Manufacturer</i>
1	1	5069 Compact I/O 16 channels AC input modules, supporting both 120 & 240 VAC signals	5069-IA16	Allen Bradley
2	2	5069 Compact I/O 8 Channel Voltage/Current Analog Input Module	5069-IF8	Allen Bradley
3	1	Controller, CompactLogix 5380, 600 KB User Memory, 8 I/Os, 16 EtherNet/IP Devices	5069-L306ER	Allen Bradley
4	4	5069 Compact I/O 16 channels non isolated normally open relay output module.	5069-OW16	Allen Bradley
5	7	5069 Compact I/O 18 pins Spring type terminal block kit in a pack of 1 pcs	5069-RTB18-SPRING	Allen Bradley
6	1	5069 Compact I/O Power terminal RTB kit for both 4 and 6 pin Spring type	5069-RTB64-SPRING	Allen Bradley
7	1	5069 Compact I/O 6 pin Spring type RTB	5069-RTB6-SPRING	Allen Bradley
8	1	Stratix 2000 Switch, Unmanaged, 5 Copper Ports	1783-US5T	Allen-Bradley
9	1	5069 Compact I/O Field Potential Distributor	5069-FPD	Allen-Bradley
10	1	UL 489, Miniature Circuit Breaker, 2 Pole, C Curve, 15 Amp	FAZ-C15/2-NA	Eaton
11	10	UL 1077, Miniature Circuit Breaker, 1 Pole, C Curve, 2 Amp	FAZ-C2/1-SP	Eaton
12	2	UL 489, Miniature Circuit Breaker, 1 Pole, C Curve, 20 Amp	FAZ-C20/1-NA-SP	Eaton
13	6	UL 1077, Miniature Circuit Breaker, 1 Pole, C Curve, 5 Amp	FAZ-C5/1-SP	Eaton
14	2	UL 489, Miniature Circuit Breaker, 3 Pole, C Curve, 6 Amp	FAZ-C6/3-NA	Eaton
15	5	Power Distribution Block, 80A, 1 Input, 6 Output, 1 Pole, DIN Rail, UD Series	UD-80A (569010)	ERIFLEX
16	16	Fuse, 2 Amp, Glass for (2002-1611)	LTF5-218002P	Little Fuse
17	18	Wire Duct Cover 3"	C3LG6-F	Panduit
18	42	Wire Duct Cover 4"	C4LG6-F	Panduit
19	18	Wire Duct, 3" x 4" p/ft	F3X4LG6	Panduit
20	42	Wire Duct, 4" x 4" p/ft	F4X4LG6	Panduit
21	1	Ground Block for 14-2 Awg Wire	LAMA2-14-QY	Panduit
22	1	Temperature Switch	3110000	Rittal
23	1	PowerPact B Circuit Breaker, 60A, 3P, 600Y/347V AC, 14kA at 600Y/347 UL EverLink	BDL36060	Schneider
24	2	TeSys D control relay - 5 NO - <= 690 V - 110 V AC standard coil	CAD50F7	Schneider
25	2	TeSys GV, top feed terminal block, for busbars, 3 P, for TeSys D and TeSys U	GV1G09	Schneider
26	1	Linery FT - Comb busbar - 63 A - 3 tap-offs - 45 mm pitch	GV2G345	Schneider

27	1	Linergy FT - Comb busbar - 63 A - 4 tap-offs - 45 mm pitch	GV2G445	Schneider
28	4	TeSys GV2 Manual Starter and Protector, thermal magnetic circuit protector, rotary handle, 4.0 to 6.3 A, screw clamp	GV2P10	Schneider
29	3	TeSys GV2 Manual Starter and Protector, thermal magnetic circuit protector, rotary handle, 6 to 10 A, screw clamp	GV2P14	Schneider
30	7	TeSys GV2 and GV3 Manual Starter and Protector, auxiliary contact block, 1 NO and 1 NC, top mount, screw clamp	GVAE11	Schneider
31	2	Safety Cover	LAD9ET1S	Schneider
32	7	TeSys D IEC contactor, 12 A, 3 P, 5 HP at 480 VAC, nonreversing, 110 VAC 50/60 Hz coil	LC1D12F7	Schneider
33	1	3 KVA 460/230 - 230/115 Transformer	3S1F	Square D
34	1	20 Pole Ground Block	PK20GTA	Square D
35	25	TOPJOB*S feedthrough terminal block; rail mount; 4-conductor; 6.2 mm wide; gray	2002-1401	WAGO
36	31	TOPJOB*S feedthrough terminal block; rail mount; 4-conductor; 6.2 mm wide; red	2002-1403	WAGO
37	20	TOPJOB*S feedthrough terminal block; rail mount; 4-conductor; 5.2 mm wide; blue	2002-1404	WAGO
38	10	TOPJOB*S ground terminal block; rail mount; 4-conductor; 5.2 mm wide; green-yellow	2002-1407	WAGO
39	3	End and intermediate plate; 1 mm thick; orange	2002-1492	WAGO
40	16	TOPJOB*S fuse disconnect terminal block; with pivoting fuse holder for 5 x 20 mm fuse; rail mount; 2-conductor; 6.2 mm wide; gray	2002-1611	WAGO
41	1	TOPJOB*S end/intermediate plate; for 2002 series fuse blocks; 2.0 mm wide; orange	2002-1692	WAGO
42	10	TOPJOB*S jumper; for 2002 series; insulated; 10-way; light-gray	2002-410	WAGO
43	3	Steel carrier rail; 35 x 7.5 mm; 1 mm thick; 2 m long; slotted; according to EN 60715; "Hole width 25 mm	210-112	WAGO
44	36	Screwless end stop; 6 mm wide; for DIN-rail 35 x 15 and 35 x 7.5; gray	249-116	WAGO
45	3	End and intermediate plate; 1 mm thick; for quadruple-deck terminal blocks; orange	280-323	WAGO
46	35	4-conductor sensor terminal block; with ground contact; with colored conductor entries; 2.5 mm ² ; CAGE CLAMP®; 2,50 mm ² ; gray	280-570	WAGO
47	2	Interface module; RJ-45; IDC technology; Cat. 6; in mounting carrier; with shield connection	289-195	WAGO
48	8	1-conductor female plug; 100% protected against mismatching; 2.5 mm ² ; Pin spacing 5 mm; 8-pole; 2,50 mm ² ; light gray	721-108/026-000	WAGO
49	3	1-conductor male connector; 100% protected against mismatching; 2.5 mm ² ; Pin spacing 5 mm; 2-pole; 2,50 mm ² ; light gray	721-602	WAGO
50	2	EPSITRON® ECO power supply; three-phase; output voltage 24 VDC; 10 A	787-740	WAGO
51	1	Redundancy Module; 2 x 9 ... 54 VDC input voltage; 2 x 12.5 A input current; 9-54 VDC output voltage; 25 A output current	787-783	WAGO
52	3	16 Port HOA Switch Board	PRO2962	WAGO

<i>Item No.</i>	<i>Qty.</i>	<i>Description</i>	<i>Part Number</i>	<i>Manufacturer</i>
1	1	PanelView 5310, 10 in. Wide Display , Touch screen, Color, 24V DC, Single Ethernet port	2713P-T10CD1	Allen Bradley
2	1	Convenience Receptacle	P-R2-K3RF0-U601	Grace
3	1	Complete fan unit for tool-free mounting on surfaces. Air throughput from 20 - 900 m³/h, protection category IP 54 as standard.	3241110	Rittal
4	1	Outlet Filters Standard	3241200	Rittal
5	1	TSFM716320XC TS 8 Flange Mount Disconnect Floor mount Enclosure	8018814	Rittal
6	1	35mm Hole Seal	OH5050C	Rittal
7	1	Handle Mechanism	9422A1	Schneider
8	1	B Frame 60" Flex Cable,	9422CSB50	Schneider
9	1	Push Button Spring Return, Recessed, Black	ZB5AA26	Schneider
10	1	Selector switch head Ø22 2-position stay put Ronis 455	ZB5AG2	Schneider
11	1	Green illuminated selector switch head Ø22 2-position stay put	ZB5AK1233	Schneider
12	3	Harmony, 22mm Push Button, XB5A operators, mounting collar for electrical blocks	ZB5AZ009	Schneider
13	3	Contact Block, Spring Clamp, Normally Open	ZBE1015	Schneider
14	1	Light Block, Spring Clamp Terminals, 110-120V, Green	ZBVG35	Schneider
15	1	WERMA audible alarm buzzer, piezoelectric, 80 dB at 1m, 2.8 kHz tone frequency, continuous tone(s), 115 VAC, permanent LED indicator(s), red, 22mm, panel mount, polycarbonate housing, red, IP65.	15010067	WERMA



SHEET #	DRAWING #	DESCRIPTION
000	2260-D-000	COVER PAGE
001	2260-D-001	DRAWING INDEX
002	2260-D-002	GENERAL ELECTRICAL SYMBOLS AND TECHNICAL NOTES
003	2260-D-003	GENERAL ELECTRICAL INSTALLATION SPECIFICATIONS AND NOTES
004	2260-D-004	480V AC POWER DISTRIBUTION
005	2260-D-005	480V AC POWER DISTRIBUTION
006	2260-D-006	480V AC POWER DISTRIBUTION
007	2260-D-007	FUTURE EXPANSION
008	2260-D-008	120V AC POWER DISTRIBUTION
009	2260-D-009	120V AC POWER DISTRIBUTION

SHEET #	DRAWING #	DESCRIPTION
010	2260-D-010	24V DC POWER DISTRIBUTION
011	2260-D-011	24V DC POWER DISTRIBUTION
012	2260-D-012	COMPACT LOGIX RACK 00 SLOTS 00 AND 01
013	2260-D-013	COMPACT LOGIX RACK 00 SLOTS 02 AND 03
014	2260-D-014	COMPACT LOGIX RACK 00 SLOTS 04 AND 05
015	2260-D-015	COMPACT LOGIX RACK 00 SLOT 06 AND FDP
016	2260-D-016	COMPACT LOGIX RACK 00 SLOT 07
017	2260-D-017	LOCAL ETHERNET NETWORK LAYOUT
018	2260-D-018	CONTROL PANEL LAYOUT
019	2260-D-019	CONTROL ENCLOSURE LAYOUT

REVISIONS:			
REV	BY	DATE	DESCRIPTION
00	BAJ	2021-03-03	RELEASE FOR CONS
DATE:		DESIGNED BY:	
2021-02-18		BA JOHNSON	
DRAWN BY:		REVIEWED BY:	
BA JOHNSON		JL ARNOLD	
UL 508A DESIGN: NO			
UL 698A DESIGN: NO			
DILLING GROUP PROJECT NUMBER:			
D215218			
CUSTOMER PROJECT/PO NUMBER:			
CR2101G 27289			
WAYNE FARMS, UNION AL			
REFRIGERATION CONTROL			
SYSTEM EVAPORATOR PANEL			
DRAWING INDEX			
DRAWING NUMBER:			
2260-D-001			
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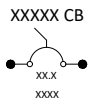
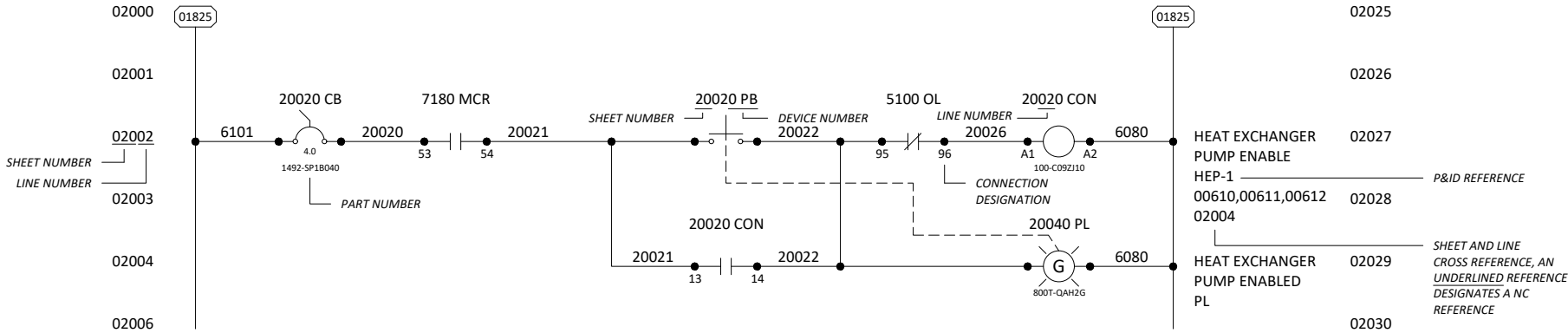


ELECTRICAL REQUIREMENTS	
SUPPLY VOLTAGE	460-480V AC, 3Ø, 60Hz
TOTAL CONNECTED Hp	13.5
TOTAL CONNECTED kVA	33.42 kVA
ESTIMATED FULL LOAD AMPS	40.2 A
MAIN FUSE/BREAKER	60 A
PLANT FUSE/BREAKER	60 A
INTERRUPT CAPACITY	6 kAIC

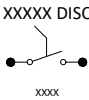
WIRE CONVERSION CHART	
METRIC	AWG
0.5	20
1.0	18
1.5	16
2.5	14
4.0	12
6.0	10
10.0	8
16.0	6
25.0	4
35.0	2
50.0	1
50.0	0

NOTES FOR INTERNAL CONTROL PANEL WIRING:

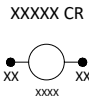
- POWER WIRING SHALL BE A MINIMUM OF #14 AWG.
- ALL 480V AC WIRING TO BE BLK.
- ALL UNGROUNDED 120V AC WIRING TO BE RED AND UTILIZE A WHT OR GRY AS THE GROUNDED NEUTRAL.
- ALL UNGROUNDED 24V DC WIRING TO BE BLU AND UTILIZE A BLU WITH WHT CHASER AS THE GROUNDED COMMON.
- ALL GROUND WIRES TO BE GRN WITH YEL TRACER.
- ALL CIRCUITS THAT MAY BE STILL ENERGIZED AFTER THE PANEL MAIN DISCONNECT IS IN THE "OFF" POSITION SHALL BE YEL.
- WIRES & MULTI-CONDUCTOR CABLES ARE TO BE SUPPORTED EVERY 3 FEET.
- ALL PLC IO TO BE TERMINATED TO A TERMINAL BLOCK.
- NO MORE THAN ONE WIRE AT EACH CONNECTION POSITION ON GROUND BARS.
- MAIN GROUND LUG TO BE LABELED AS "PE".
- ALL ENCLOSURE DOORS ARE TO HAVE A GROUND BONDING WIRE.
- THERE IS TO BE NO MORE THAN 2 WIRES ON ANY TERMINATION POINT. IF 2 WIRES ARE WIRED TO THE SAME CONNECTION POINT, THE TERMINAL RATING MUST MATCH OR EXCEED THE THE COMBINED GAUGE OF THE 2 WIRES.
- ALL DEVICES SHALL BE TOUCH SAFE, HAVE MOLDED COVERS OR LEXAN COVERS. IF LEXAN COVERS ARE USED, TEST HOLES MUST BE DRILLED TO ALLOW FOR TESTING OF VOLTAGES WITH MULTI-METER.
- HIGH VOLTAGE POWER DISTRIBUTION BLOCKS SHALL ALSO BE LABELED AS SUCH.
- ALL WIRES SHALL HAVE WIRE LABELS AT ALL TERMINATION POINTS.
- ALL DEVICES SHALL MATCH THE MANUFACTURERS TORQUE RECOMMENDATIONS AND BE PERMANENTLY MARKED ON THE BACK PANEL.
- ALL DEVICES SHALL BE MARKED ACCORDING TO THE DESIGNATION PROVIDED ON THE ELECTRICAL SCHEMATICS.



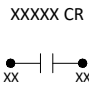
CIRCUIT BREAKER



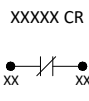
DISCONNECT SWITCH
NORMALLY OPEN (NO)
HELD CLOSED



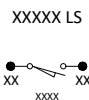
COIL
(CONTROL RELAYS,
CONTACTORS, TIMER, ETC.)



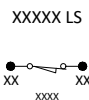
DRY CONTACT
NORMALLY OPEN (NO)



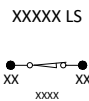
DRY CONTACT
NORMALLY CLOSED (NC)



LIMIT SWITCH
NORMALLY OPEN (NO)



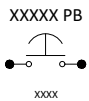
LIMIT SWITCH
NORMALLY OPEN (NO)
HELD CLOSED



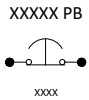
LIMIT SWITCH
NORMALLY CLOSED (NC)



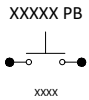
LIMIT SWITCH
NORMALLY CLOSED (NC)
HELD OPEN



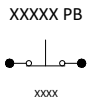
MUSHROOM HEAD
PUSHBUTTON
NORMALLY OPEN (NO)



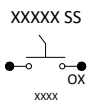
MUSHROOM HEAD
PUSHBUTTON
NORMALLY CLOSED (NC)



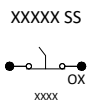
PUSHBUTTON
NORMALLY OPEN (NO)



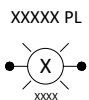
PUSHBUTTON
NORMALLY CLOSED (NC)



SELECTOR SWITCH
NORMALLY OPEN (NO)



SELECTOR SWITCH
NORMALLY CLOSED (NC)



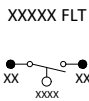
INDICATOR LIGHT
(PILOT LIGHT, STACK LIGHT,
ETC.)



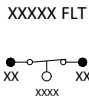
SOLENOID



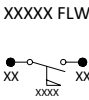
LIGHT
(LED LIGHT BAR, FLORESCENT
LIGHT, ETC.)



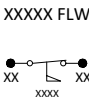
FLOAT SWITCH
NORMALLY OPEN (NO)



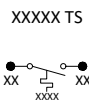
FLOAT SWITCH
NORMALLY CLOSED (NC)



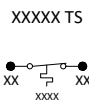
FLOW SWITCH
NORMALLY OPEN (NO)



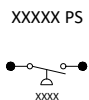
FLOW SWITCH
NORMALLY CLOSED (NC)



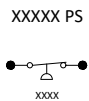
TEMPERATURE SWITCH
NORMALLY OPEN (NO)



TEMPERATURE SWITCH
NORMALLY CLOSED (NC)



PRESSURE SWITCH
NORMALLY OPEN (NO)



PRESSURE SWITCH
NORMALLY CLOSED (NC)



POWER RECEPTACLE

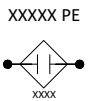


PHOTO EYE OR
REED SWITCH 2 WIRE
NORMALLY OPEN (NO)

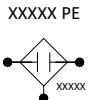


PHOTO EYE OR
REED SWITCH 3 WIRE
NORMALLY OPEN (NO)

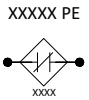


PHOTO EYE OR
REED SWITCH 2 WIRE
NORMALLY CLOSED (NC)

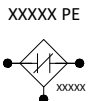
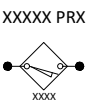
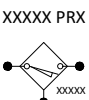


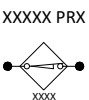
PHOTO EYE OR
REED SWITCH 3 WIRE
NORMALLY CLOSED (NC)



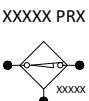
PROXIMITY SWITCH
2 WIRE
NORMALLY OPEN (NO)



PROXIMITY SWITCH
3 WIRE
NORMALLY OPEN (NO)



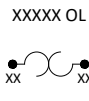
PROXIMITY SWITCH
2 WIRE
NORMALLY CLOSED (NC)



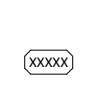
PROXIMITY SWITCH
3 WIRE
NORMALLY CLOSED (NC)



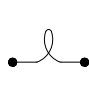
FUSE



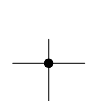
MOTOR OVERLOAD



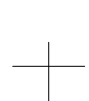
LINE REFERENCE



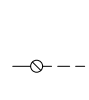
JUMPER



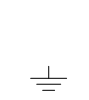
WIRE CROSS
WITH CONNECTION



WIRE CROSS
WITHOUT CONNECTION



TERMINAL WITH
FIELD WIRING DESIGNATION



GROUND CONNECTION



PROTECTED EARTH
GROUND



CUSTOMER SUPPLIED
OR CONTINUANCE
REFERENCE

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REVISIONS:			
REV	BY	DATE	DESCRIPTION
00	BAJ	2021-03-03	RELEASE FOR CONST.

DATE: 2021-02-18	DESIGNED BY: BA JOHNSON
DRAWN BY: BA JOHNSON	REVIEWED BY: JL ARNOLD

UL 508A DESIGN: NO
UL 698A DESIGN: NO
DILLING GROUP PROJECT NUMBER: D215218
CUSTOMER PROJECT/PO NUMBER: CR2101G 27289

WAYNE FARMS, UNION AL
REFRIGERATION CONTROL
SYSTEM EVAPORATOR PANEL

GENERAL
ELECTRICAL SYMBOLS
AND
TECHNICAL NOTES

DRAWING NUMBER:
2260-D-002

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C:\Users\brad.phinson\Desktop\DD215218 - REPUBLIC - REFRIGERATION - Wayne Farms Evaporator Control Panel, Union Springs, AL - Job D215218 - 2021-03-03.dwg 2021-03-03 08:28AM C:\Users\brad.phinson\Desktop\DD215218 - REPUBLIC - REFRIGERATION - Wayne Farms Evaporator Control Panel, Union Springs, AL - Job D215218 - 2021-03-03.dwg

1 BASE SPECIFICATION

- ## 2 GENERAL REQUIREMENTS

- 2.10 FINAL OPERATIONS
- 2.10.1 UPON COMPLETION OF THE WORK, ALL ELECTRICAL APPARATUS SHALL BE CLEANED OF DUST AND DEBRIS SUCH THAT THE FINAL CONDITION SHALL BE ACCEPTABLE TO BOTH THE ENGINEER AND OWNER.
- 2.10.2 WHERE PAINTED SURFACES OF THE EQUIPMENT HAVE BEEN ABUSED OR RUSTED DURING CONSTRUCTION THE ELECTRICAL CONTRACTOR SHALL PAINT SAME TO MATCH ORIGINAL FACTORY OR SURROUNDING FINISH.

3 CONTRACT DOCUMENTS

- 3.1 THE WORK SHALL BE IN ACCORDANCE WITH THE DRAWINGS AS NECESSARY TO OBTAIN A COMPLETELY SATISFACTORY AND OPERABLE INSTALLATION. DRAWINGS PERTAINING TO OTHER TRADES SHALL BE REVIEWED AND WORK SHALL BE COORDINATED TO PREVENT PHYSICAL INTERFERENCES.
- 4 SYSTEM VOLTAGES
- 4.1 THE VOLTAGES OF THE VARIOUS PORTIONS OF THE ELECTRICAL SYSTEM ARE AS FOLLOWS:
- POWER UTILIZATION: 480V AC, 3Ø, 3 WIRE, 60Hz
 - CONVENIENCE RECEPTACLES AND LIGHTING: 120/240V AC, 1Ø
 - GROUNDED CONTROL CIRCUITS: 120V AC, 1Ø AND 24V DC

5 CONDUIT WORK

- 5.1 ALL CONDUIT SHALL BE 3/4" MINIMUM OR OTHERWISE NOTED ON THE DRAWINGS. THE TYPE SHALL FIT THE ENVIRONMENTAL REQUIREMENTS OF THE SPACE IN WHICH IT IS INSTALLED.
- 5.2 CONDUIT SHALL BE INSTALLED IN STRICT ACCORDANCE WITH THE REQUIREMENTS OF THE NATIONAL ELECTRICAL CODE AND THE FOLLOWING:
- 5.2.1 CONDUIT SHALL BE PROPERLY SECURED IN PLACE WITH APPROVED FASTENERS. WOODEN PLUGS INSERTED IN MASONRY OR CONCRETE SHALL NOT BE USED AS A BASE TO SECURE CONDUIT SUPPORTS. CONDUIT FASTENERS SHALL BE SECURED BY TOGGLE BOLTS ON HOLLOW MASONRY, EXPANSION SHIELDS IN CONCRETE OR BRICK, MACHINE SCREWS AND NUTS ON METAL SURFACE, AND WOOD SCREWS ON WOOD CONSTRUCTION. GALVANIZED ONE HOLE MALLEABLE IRON CONDUIT CLAMPS WITH BACKS SHALL BE USED FOR ALL CONTACT WITH BUILDING SURFACES. ANGLE IRON RACKS OR APPROVED EQUIVALENT SHALL BE PROVIDED FOR ALL MULTIPLE ANGLE CONDUIT RUNS. THE RACKS SHALL BE OF WELDED CONSTRUCTION AND SHALL UTILIZE IRON SUPPORTS OR EQUIVALENT (NO RODS PERMITTED) SECURELY ANCHORED TO THE BUILDING STRUCTURE OR WELDED TO BUILDING STEEL. WELDS TO BUILDING STEEL SHALL BE PARALLEL TO THE WEB. IF USED, ANGLE IRON SHALL BE SIZED AS REQUIRED, 2" X 2" X 3/16" MINIMUM. RACKS SHALL PROVIDE A MINIMUM OF 25 PERCENT SPARE CAPACITY FOR FUTURE CONDUITS. CONDUITS INSTALLED ON RACKS SHALL BE ANCHORED WITH CONDUIT CLAMPS. FERROUS METAL RACKS SHALL BE PAINTED WITH A PRIMER COAT BEFORE CONDUITS ARE INSTALLED.
- 5.2.2 ALL HORIZONTAL EXPOSED CONDUIT RUNS SHALL BE LEVEL, STRAIGHT, AND PARALLEL TO WALLS AND FLOORS. ALL VERTICAL CONDUIT RUNS SHALL BE PLUMB. OPEN CONDUIT SHALL BE SEALED WITH CONDUIT CLOSURES DURING COURSE OF CONSTRUCTION.
- 5.2.3 CONDUITS SHALL NOT BE SECURED TO OR COME IN CONTACT WITH PIPES. PIPE FLANGE BOLTS AND EQUIPMENT BOLTS SHALL NOT BE USED TO ANCHOR CONDUITS OR CONDUIT SUPPORTS. CONDUITS SHALL NOT BE SUPPORTED BY PIPING OR OTHER CONDUITS. CONDUITS SHALL CLEAR HEATED SURFACES, HEATED PIPES AND INSULATION BY AT LEAST SIX INCHES.

- 5.2.4 EXPANSION FITTINGS SHALL BE INSTALLED ACROSS EXPANSION JOINTS IN STRUCTURES AND CONCRETE CONSTRUCTION WHERE SUCH JOINTS ARE SHOWN ON THE ARCHITECTURAL AND STRUCTURAL DRAWINGS, OR AS REVIEWED BY THE ENGINEER.
- 5.2.5 SPARE CONDUITS OR THOSE INDICATED FOR "FUTURE" USE SHALL BE CLOSED BY THREADED PLUGS OR CAPS.
- 5.2.6 ALL RACEWAY SYSTEMS SHALL BE CLEANED OF DEBRIS AND MOISTURE PRIOR TO INSTALLING ELECTRICAL CONDUCTORS.
- 5.2.7 WHERE CONDUIT PASSES BETWEEN AREAS SUBJECT TO DIFFERENT TEMPERATURES, THE CONDUIT SHALL BE SEALED TO PREVENT INTERCHANGE OF AIR AND THE FORMATION OF CONDENSATION. THIS SHALL BE DONE BY THE USE OF A CONDUIT FITTING AND DUXSEAL OR OTHER APPROVED MASTIC MATERIAL.
- 5.2.8 ALL CONDUIT SHALL HAVE A SEPARATE GROUND WIRE (GREEN) CONNECTED TO TRUE EARTH GROUND.

6 BOXES AND ENCLOSURES

- 6.1 JUNCTION BOXES, PULL BOXES, AND TERMINAL BOXES SHALL BE PROVIDED WHERE INDICATED ON THE DRAWINGS. ALL BOXES SHALL CONFORM TO THE UL/NEMA TYPE DESIGNATION FOR THE AREA IN WHICH THEY ARE INSTALLED. BOXES SHALL BE ACCURATELY SET AND SECURELY HELD IN PLACE.
- 6.2 TERMINAL BOXES SHALL BE FURNISHED WITH SUBPANELS AND TERMINAL BLOCKS. TERMINAL BLOCKS SHALL BE 600-VOLT, CHANNEL MOUNTED TUBULAR SCREW TYPE OR SPRING CLAMP WITH PRESSURE PLATE, VINYL MARKINGS STRIPS, END BARRIERS, AND END ANCHORS. A MINIMUM OF 20% SPARE TERMINALS SHALL BE PROVIDED. WIRE OR TERMINAL NUMBERS. AS SHOWN ON THE DRAWINGS, SHALL BE PUT ON THE MARKING STRIPS IN A NEAT AND LEGIBLE MANNER. PULL NUMBERS ARE NOT ACCEPTABLE.

7 RACKS AND SUPPORTS

- 7.1 RACKS AND SUPPORTS FOR CONTROL PANELS, STARTERS, CONTROL STATIONS, ETC., SHALL BE FURNISHED AND INSTALLED AND SHALL BE CONSTRUCTED OF ANGLE IRON OR MATERIALS SHOWN ON DRAWINGS. ALL WALL MOUNTED EQUIPMENT SHALL BE SUPPORTED BY STRUCTURAL CHANNEL, MOUNTED VERTICALLY, TO PROVIDE TWO-INCH MINIMUM SPACE BETWEEN THE EQUIPMENT AND THE WALL. STRUT TYPE CHANNEL SHALL NOT BE INSTALLED UNLESS SPECIFICALLY SPECIFIED ON THE DRAWINGS. RACKS, EXCEPT THOSE CONSTRUCTED OF STAINLESS STEEL, SHALL BE PAINTED WITH A PRIMER COAT BEFORE EQUIPMENT IS MOUNTED. ANY SUBSEQUENT FIELD WELDS SHALL BE CLEANED AND PRIMED.

8 WIRE AND CABLE

- 8.1 WIRE AND CABLE SHALL BE FURNISHED AND INSTALLED UNDER THIS DIVISION. CONDUCTORS SHALL BE STRANDED COPPER.
- 8.2 600-VOLT POWER WIRE AND CABLE
 - 8.2.1 POWER WIRE AND CABLE SHALL BE 600-VOLT, TYPE THHN/THWN UNLESS SPECIFIED OTHERWISE. WHEN USING THHN/THWN, CONDUITS SHALL BE SIZED FOR 30% OR LESS FILL.
 - 8.2.2 WIRE FOR 120/240-VOLT CIRCUITS SHALL BE COLOR-CODED BLACK AND RED FOR HOT, WHITE FOR NEUTRAL, AND GREEN FOR GROUND.
- 8.3 CONTROL WIRE
 - 8.3.1 CONTROL WIRE SHALL BE 600-VOLT SINGLE CONDUCTOR TYPE THHN/THWN SEVEN STRAND MINIMUM, #14 AWG MINIMUM, COLOR-CODED, RED FOR HOT AND WHITE FOR NEUTRAL OR OTHERWISE NOTED ON THE DRAWINGS.
 - 8.3.2 THE MINIMUM BENDING RADIUS AND MAXIMUM PULLING PRESSURES AND TENSIONS OF LOW LEVEL, MULTI-CONDUCTOR SHIELDED OR UNSHIELDED CABLE SHALL NOT EXCEED MANUFACTURER'S RECOMMENDATION.
- 8.4 INSTRUMENTATION WIRING
 - 8.4.1 INSTRUMENTATION CABLES SHALL BE SHIELDED TWISTED PAIR, #18 AWG WITH PVC JACKET, AND RUN IN SEPARATE CONDUITS. CABLE SHALL HAVE AN OVERALL FOIL SHIELD WITH DRAIN WIRE.
 - 8.4.2 CABLES SHALL BE CONTINUOUS BETWEEN TERMINATION POINTS AND SHALL NOT BE SPLICED.
 - 8.4.3 CABLES SHALL BE MANUFACTURED BY ALPHA, BELDEN, OR CARROL
- 8.5 SPLICES AND TERMINATIONS
 - 8.5.1 POWER WIRE SHALL BE SPLICED AND TERMINATED USING TERMINAL LUGS, BRASS NUTS, BOLTS, AND WASHERS WITH FIVE LAYER OF MINNESOTA MINING SCOTCH #88 INSULATING TAPE APPLIED. WIRE NUTS, SCOTCHLOCKS, AND SIMILAR DEVICES SHALL NOT BE USED. TERMINAL LUGS SHALL BE THOMAS & BETTS STA-KON OR AMP CRIMP-ON TYPE. MOTOR T LEADS SHALL BE BLACK.

- 8.5.2 CONTROL WIRE SHALL NOT BE SPICED UNLESS A COMMON WIRE IS USED FOR SEVERAL SOLENOIDS, PRESSURE SWITCHES, ETC., THESE SPLICES SHALL BE MADE IN AN APPROPRIATE J BOX.
- 8.6 EACH CONDUCTOR SHALL BE MARKED WITH EITHER THE WIRE NUMBER OR TERMINAL NUMBER AT EACH TERMINATION POINT, AS SPECIFICALLY DESIGNATED ON THE DRAWINGS. PULL NUMBERS ARE NOT ACCEPTABLE. BRADY WIRE MARKERS SHALL BE USED.
- 8.7 WIRE AND CABLE TESTING
- 8.7.1 THE INSULATION RESISTANCE OF ALL FEEDERS TO CONTROL PANELS AND MOTOR LEADS LEAVING THE CONTROL PANELS SHALL BE PROOF-TESTED BY DIRECT CURRENT OVERPOTENTIAL TESTING AFTER INSTALLATION AND BEFORE BEING ENERGIZED. TEST VOLTAGES AND PERIODS SHALL BE IN ACCORDANCE WITH THE MANUFACTURER'S RECOMMENDATIONS.
- 8.7.2 A RECORD OF TESTS SHALL BE MAINTAINED AND PRESENTED TO THE ENGINEER UPON COMPLETION. ALL TESTS SHALL BE WITNESSED BY THE ENGINEER PER THEIR DISCRETION.

9 CONTROLS AND INSTRUMENTS

- 9.1 THE CONTRACTOR SHALL FURNISH AND/OR INSTALL ALL CONTROLS AND INSTRUMENTS IN ACCORDANCE WITH THE DRAWINGS. ADJUSTMENT, SETTING, AND CALIBRATION SHALL BE MADE AS DIRECTED BY THE ENGINEER.

10 TESTING

- 10.1 MAKE TESTS OR FURNISH PERSONNEL TO ASSIST IN MAKING TESTS TO ASSURE ALL EQUIPMENT, ITS CONTROL AND SEQUENCE OF OPERATION IS IN ACCORDANCE WITH THE DRAWINGS AND/OR MANUFACTURER'S INSTRUCTIONS. ALL TESTS SHALL BE WITNESSED BY THE ENGINEER.

11 EQUIPMENT GROUNDING

- 11.1 ALL EQUIPMENT SHALL BE GROUNDED AS SHOWN AND IN ACCORDANCE WITH THE NATIONAL ELECTRIC CODE (NEC).

12 EQUIPMENT LABELING

- 12.1 ENGRAVED NAMEPLATES SHALL BE FURNISHED FOR ALL ITEMS SUCH AS CONTROL PANELS, MOTOR STARTERS, AND ALARM PANELS. NAMEPLATES SHALL BE THREE-PLY PHENOLIC OR VINYL MATERIAL WITH WHITE SURFACE AND BLACK INNER CORE. NAMEPLATES SHALL BE ATTACHED WITH FOUR DRIVE SCREWS. SECURING WITH ADHESIVES IS NOT ACCEPTABLE. ANY CONTROL PANELS HAVING MULTIPLE POWER SUPPLIES SHALL HAVE THE FOLLOWING CAUTION TAG AFFIXED TO THEM "CAUTION VOLTAGE WIRES REPRESENT FOREIGN VOLTAGE PRESENT IN THIS PANEL". THIS TAG IS TO BE MADE WITH YELLOW TAPVOPLY WITH 1/8 INCH HIGH BLACK LETTERS, OR EQUAL.

13 OVERLOAD ELEMENTS

- 13.1 CONTRACTOR SHALL FURNISH AND INSTALL THERMAL OVERLOAD ELEMENTS FOR NEMA TYPE STARTERS. OVERLOAD ELEMENTS SHALL BE SIZED BASED ON THE MOTOR NAME PLATE FULL LOAD AMPS.
- 13.2 CONTRACTOR SHALL ADJUST ELECTRONIC OVERLOAD ELEMENTS FOR IEC TYPE CONTACTORS. OVERLOAD ELEMENTS SHALL BE SIZED BASED ON THE MOTOR NAME PLATE FULL LOAD AMPS.

14 DOCUMENTATION

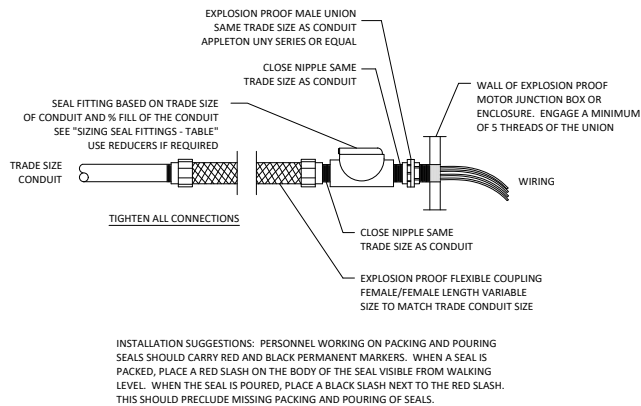
- 14.1 CONTRACTOR SHALL MAINTAIN AND DELIVER DRAWINGS WITH ANY MARK-UPS, CHANGES, MODIFICATIONS OR CORRECTIONS. DRAWINGS SHALL BE SUBMITTED TO THE ENGINEER FOR CONSTRUCTION RECORD PURPOSES.

15 HAZARDOUS AREAS

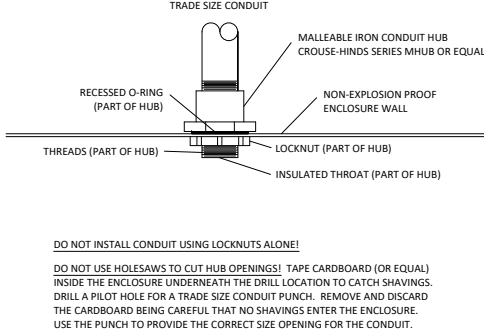
- 15.1 THE DRAWINGS DO NOT SHOW "SEALING FITTINGS" OR OTHER SPECIFIC "CODE REQUIREMENTS FOR ELECTRICAL INSTALLATION" IN HAZARDOUS AREAS. ALL MATERIAL AND EQUIPMENT SHALL BE FURNISHED AND INSTALLED IN ACCORDANCE WITH THE REQUIREMENTS OF THE NEC. EXAMPLES OF THESE REQUIREMENTS ARE DISPLAYED BELOW.

16 LIABILITY

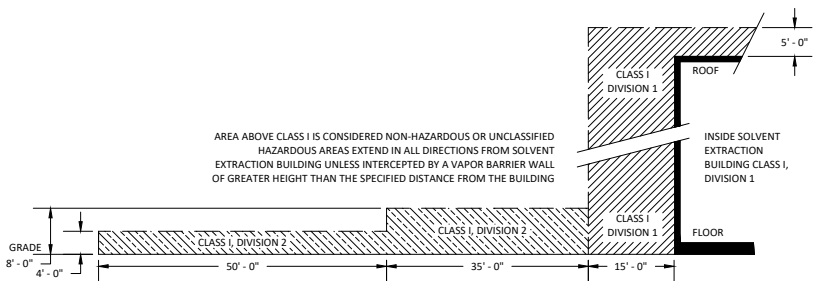
- 15.1 NO MATTER WHAT INDICATION IS PLACED ON THIS SET OF PRINTS, IF SOMETHING INDICATED DOES NOT CONFORM TO LOCAL, NATIONAL OR INTERNATIONAL CODE AS APPLICABLE, THE CONTRACTOR MUST BRING THIS TO THE ATTENTION OF THE ENGINEER. SPECIFICATIONS CONTAINED IN THIS DRAWING PACKAGE MUST NOT BE FOLLOWED BLINDLY AND THE CONTRACTOR IS REQUIRED TO ADHERE TO ALL GOVERNING BODIES EFFECTED BY THIS SYSTEM.



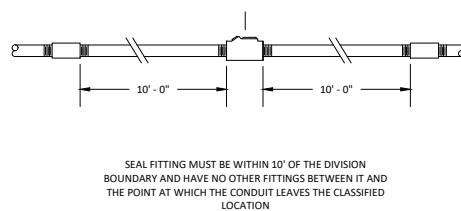
TYPICAL INSTALLATION FOR SEAL FITTING AT NEMA 7 ENCLOSURE



TYPICAL CONDUIT ENTRY INTO A NON-EXPLOSION PROOF ENCLOSURE



EXAMPLE: CLASSIFICATION OF HAZARDOUS AREAS SURROUNDING A SOLVENT EXTRACTION BUILDING EXCERPTED FROM NFPA 36



LOCATION AND INSTALLATION OF BOUNDARY SEAL FITTINGS

SIZING SEAL FITTINGS - TABLE

THE 2013 NEC STATES IN ARTICLE 501.5(C)(6) CONDUIT FILL THAT "THE CROSS-SECTIONAL AREA OF THE CONDUIT PERMITTED IN A SEAL SHALL NOT EXCEED 25% OF THE CROSS-SECTIONAL AREA OF A RIGID METAL CONDUIT OF THE SAME TRADE SIZE UNLESS SPECIFICALLY IDENTIFIED FOR A HIGHER PERCENTAGE OF FILL." THE TABLE BELOW LISTS THE APPLICABLE SIZES FOR STANDARD AND EXPANDED SEALS BASED ON THE TRADE SIZE OF THE CONDUIT USED AND THE PERCENT FILL ASSUMING THAT THE SEAL IS NOT IDENTIFIED FOR GREATER THAN 25% FILL.

STANDARD SEALS TO BE APPLETON SERIES EY (VERTICAL/HORIZONTAL) OR EQUAL. EXPANDED SEALS TO BE O-2/GEDNEY SERIES EYAK (VERTICAL/HORIZONTAL) OR EQUAL.

TRADE SIZE CONDUIT (IN INCHES)	CONDUIT CROSS SECTIONAL AREA (IN INCHES ²)	25% OF CONDUIT CROSS SECTIONAL AREA (IN INCHES ²)	TRADE SIZE SEAL REQUIRED BY CONDUIT FILL * (SEAL IN INCHES)			
			25%	30%**	35%**	40%**
0.50	0.314	0.07850	0.50	0.75	0.75	0.75
0.75	0.424	0.10600	0.75	1.00	1.00	1.00
1.00	0.887	0.22175	1.00	1.25	1.25	1.25
1.25	1.026	0.31850	1.25	1.50	1.50	2.00
1.50	1.571	0.51775	1.50	2.00	2.00	2.00
2.00	3.408	0.85200	2.00	2.50	3.50	3.00
2.50	4.866	1.21650	2.50	3.00	3.00	3.50
3.00	7.499	1.87475	3.00	3.50	4.00	4.00
3.50	10.150	2.53750	2.50	4.00	EXPANDED 3.50	EXPANDED 3.50
4.00	12.882	3.22050	4.00	EXPANDED 4.00	EXPANDED 4.00	EXPANDED 4.00
5.00	20.212	5.05300	NOT AVAILABLE	NOT AVAILABLE	NOT AVAILABLE	NOT AVAILABLE
6.00	29.158	7.28950	NOT AVAILABLE	NOT AVAILABLE	NOT AVAILABLE	NOT AVAILABLE

* SEE CONDUIT AND WIRE SCHEDULE FOR CONDUIT FILL

** USE THREAD RIGID CONDUIT SEAL REDUCING BUSHINGS (STEEL OR MALLEABLE IRON) FOR CONDUIT-TO-STANDARD.

ALL SEALS LISTED ABOVE ARE STANDARD SEALS EXCEPT THOSE MARKED AS "EXPANDED 25%" OR "EXPANDED". THE EXPANDED VERSION IS A SPECIAL SEAL, WITH AN INCREASED INTERIOR CROSS-SECTIONAL AREA TO CONFORM TO THE CODE REQUIREMENT FOR UP TO 40% CONDUIT FILL. EXPANDED SEALS HAVE HUBS DESIGNED FOR THE REFERENCED TRADE SIZE CONDUIT AND WILL NOT REQUIRE REDUCING BUSHINGS. EXPANDED VERSION SEALS OF THE APPROPRIATE TRADE SIZE MAY BE SUBSTITUTED FOR STANDARD SEALS AT ANY CONDUIT FILL PERCENTAGE.

2013 NEC, 501.16(A)(1), SEALS ARE REQUIRED WHERE A CONDUIT ENTERS INTO AN EXPLOSION PROOF ENCLOSURE WHERE EITHER OF THE FOLLOWING CONDITIONS APPLY:

- THE ENCLOSURE OR THE APPARATUS, SUCH AS SWITCHES, CIRCUIT BREAKERS, FUSES, RELAYS OR RESISTORS, THAT MAY PRODUCE ARC, SPARKS, OR HIGH TEMPERATURES THAT ARE CONSIDERED TO BE AN IGNITION SOURCE IN NORMAL OPERATION.
- THE ENTRY IS TRADE SIZE 2" OR LARGER AND THE ENCLOSURE CONTAINS TERMINALS, SPLICES OR TAPS.

2013 NEC, 501.15(A)(4). IN EACH CONDUIT RUN LEAVING A CLASS 1, DIVISION 1 LOCATION, A SEALING FITTING IS REQUIRED AT THE DIVISION BOUNDARY. THE SEALING FITTING SHALL BE PERMITTED ON EITHER SIDE OF THE BOUNDARY OF SUCH LOCATION WITHIN 10' OF THE BOUNDARY AND SHALL BE DESIGNED AND INSTALLED SO AS TO MINIMIZE THE AMOUNT OF GAS OR VAPOR WITHIN THE DIVISION 1 LOCATION. IF THE CONDUIT FROM BEING CONNECTED TO THE CONDUIT BEYOND THE SEAL, EXCEPT FOR LISTED EXPLOSION PROOF REDUCERS, AT THE CONDUIT SEAL, THERE SHALL BE NO UNION, COUPLING, BOP OR FITTING BETWEEN THE CONDUIT SEAL AND THE POINT AT WHICH THE CONDUIT LEAVES THE DIVISION 1 LOCATION.

2013 NEC, 501.15(B)(2). IN EACH CONDUIT RUN PASSING FROM CLASS 1, DIVISION 2 LOCATION, A SEAL FITTING IS REQUIRED WHERE THE CONDUIT PASSES INTO A NON-CLASSIFIED AREA. SEAL LOCATIONS AT THE BOUNDARY ARE THE SAME AS SPECIFIED IN 2013 NEC, 501.15(A)(4) ABOVE.

2013 NEC, 504.70. SEALS FOR INTRINSICALLY SAFE (IS) CONDUITS. CONDUITS AND CABLES THAT ARE REQUIRED TO BE SEALED BY 501.15, 502.15. SHALL BE SEALED TO MINIMIZE THE PASSAGE OF GASES, VAPORS OR DUST. EXCEPTION: SEALS SHALL NOT BE REQUIRED FOR ENCLOSURES THAT CONTAIN ONLY INTRINSICALLY SAFE EQUIPMENT, AS REQUIRED BY 501.15(E)(4). THIS INCLUDES CABLES THAT DO NOT HAVE GAS/VAPOR/IGHT CONTINUOUS SEALS. SEALS SHALL BE SEALED AT THE BOUNDARY OF THE CLASSIFIED AND UNCLASSIFIED LOCATIONS TO PREVENT PASSAGE OF GASES AND VAPORS INTO AN UNCLASSIFIED LOCATION.

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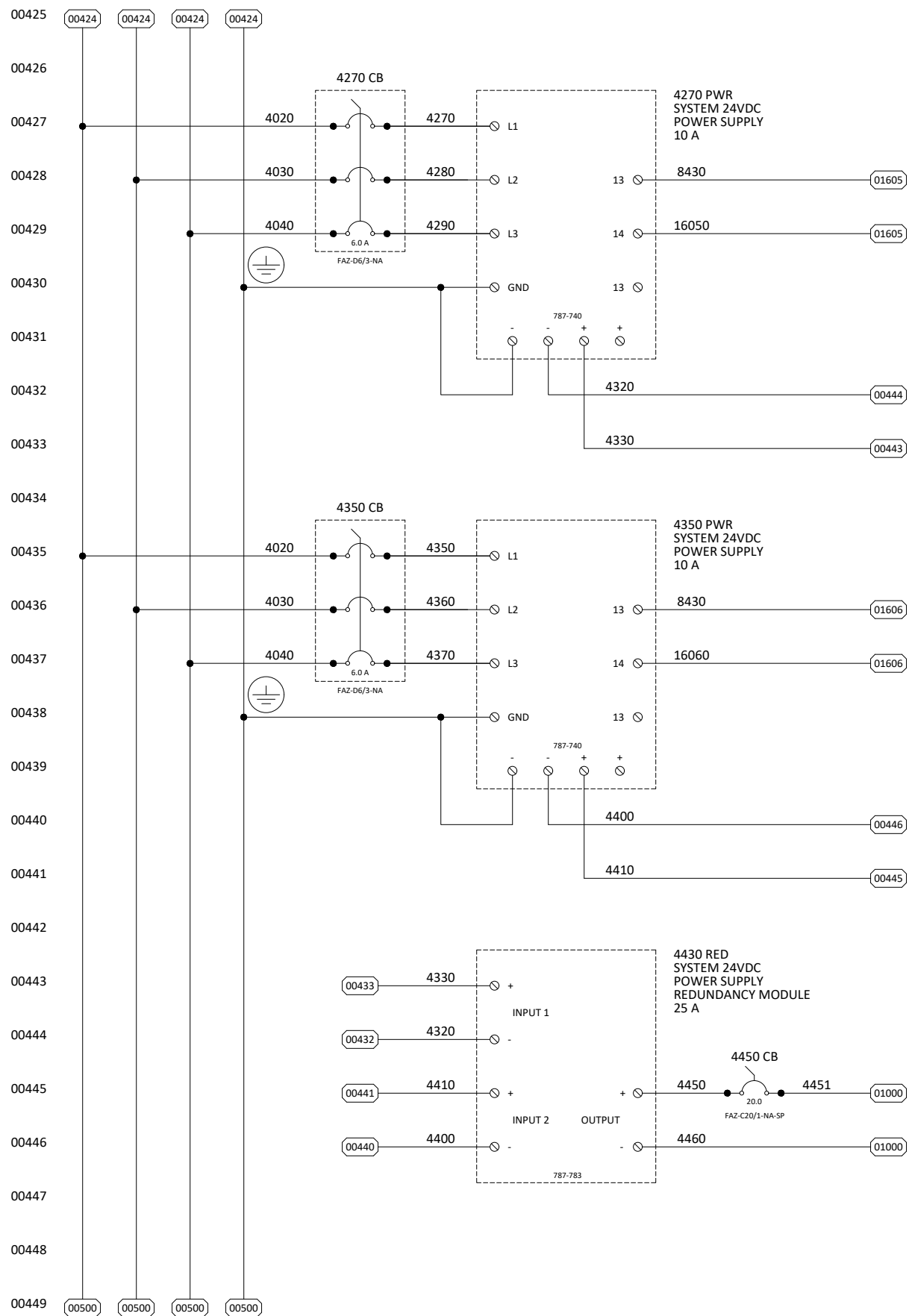
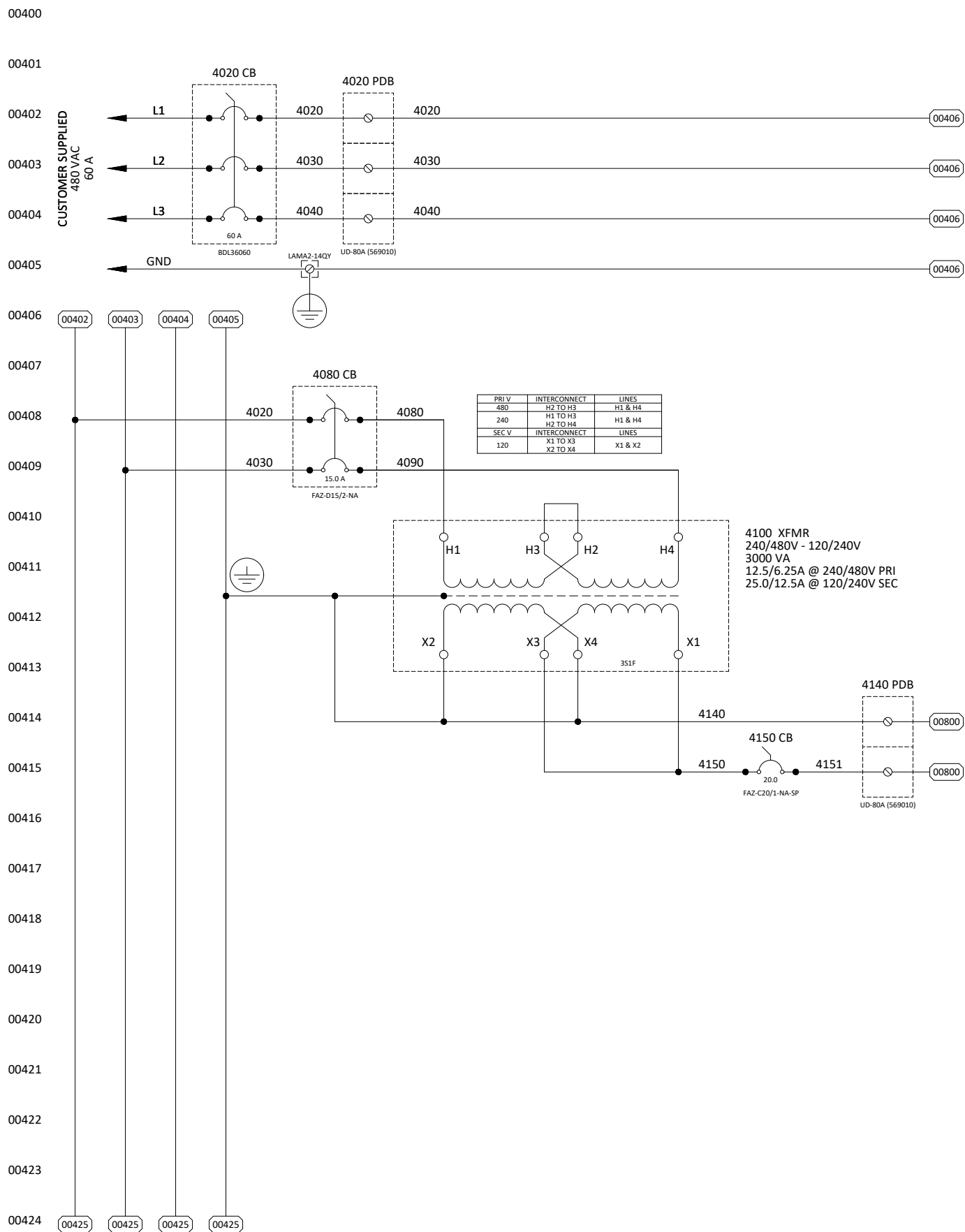
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DILLING GROUP PROJECT NUMBER:
D215218
CUSTOMER PROJECT/PO NUMBER:
CR2101G 27289
WAYNE FARMS, UNION AL
REFRIGERATION CONTROL
SYSTEM EVAPORATOR PANEL

GENERAL ELECTRICAL INSTALLATION SPECIFICATIONS AND NOTES

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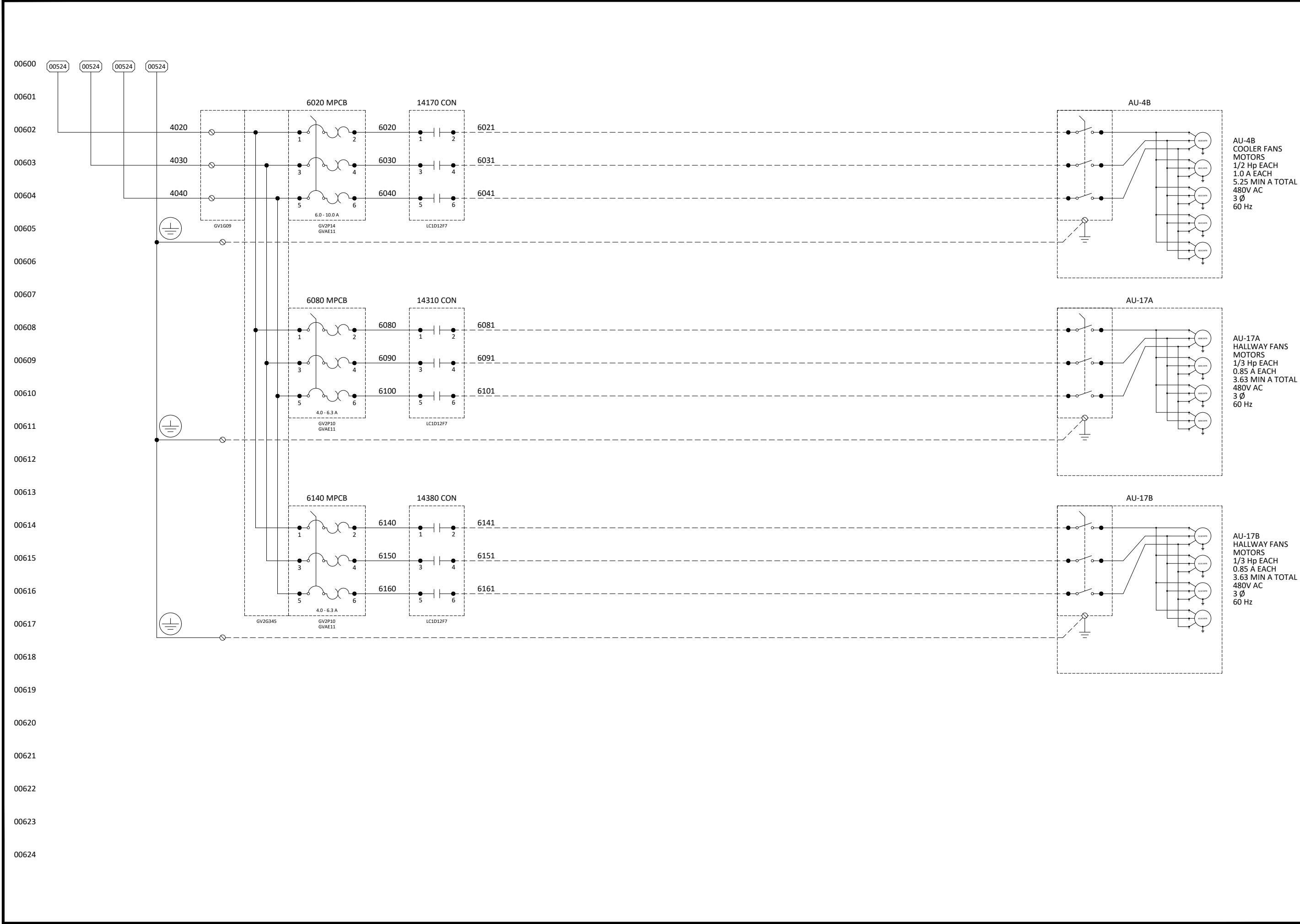
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AU-4B
COOLER FANS
MOTORS
1/2 Hp EACH
1.0 A EACH
5.25 MIN A TOTAL
480V AC
3 Ø
60 Hz

AU-17A
HALLWAY FANS
MOTORS
1/3 Hp EACH
0.85 A EACH
3.63 MIN A TOTAL
480V AC
3 Ø
60 Hz

AU-17B
HALLWAY FANS
MOTORS
1/3 Hp EACH
0.85 A EACH
3.63 MIN A TOTAL
480V AC
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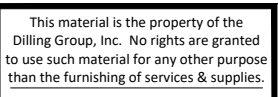
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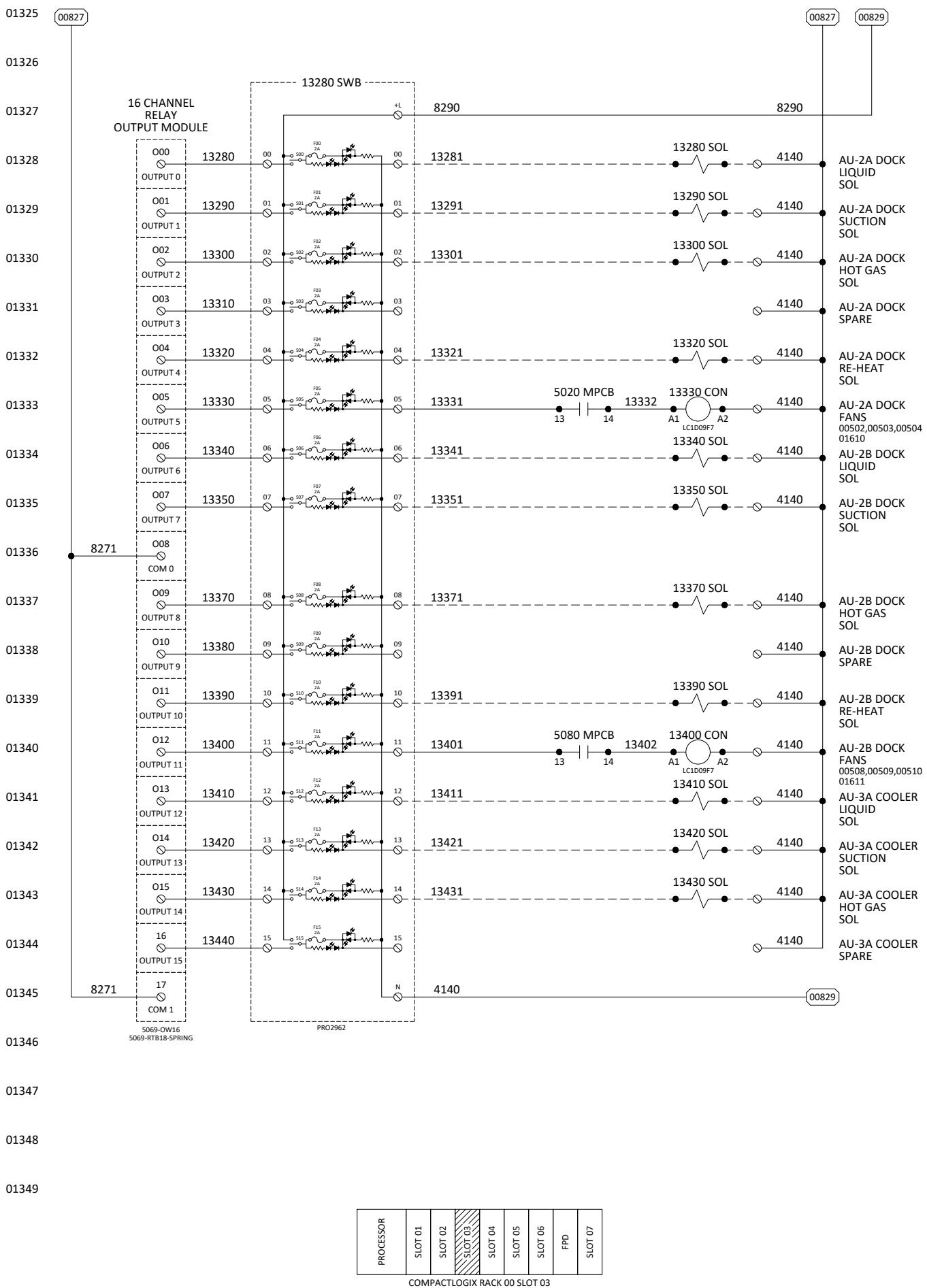
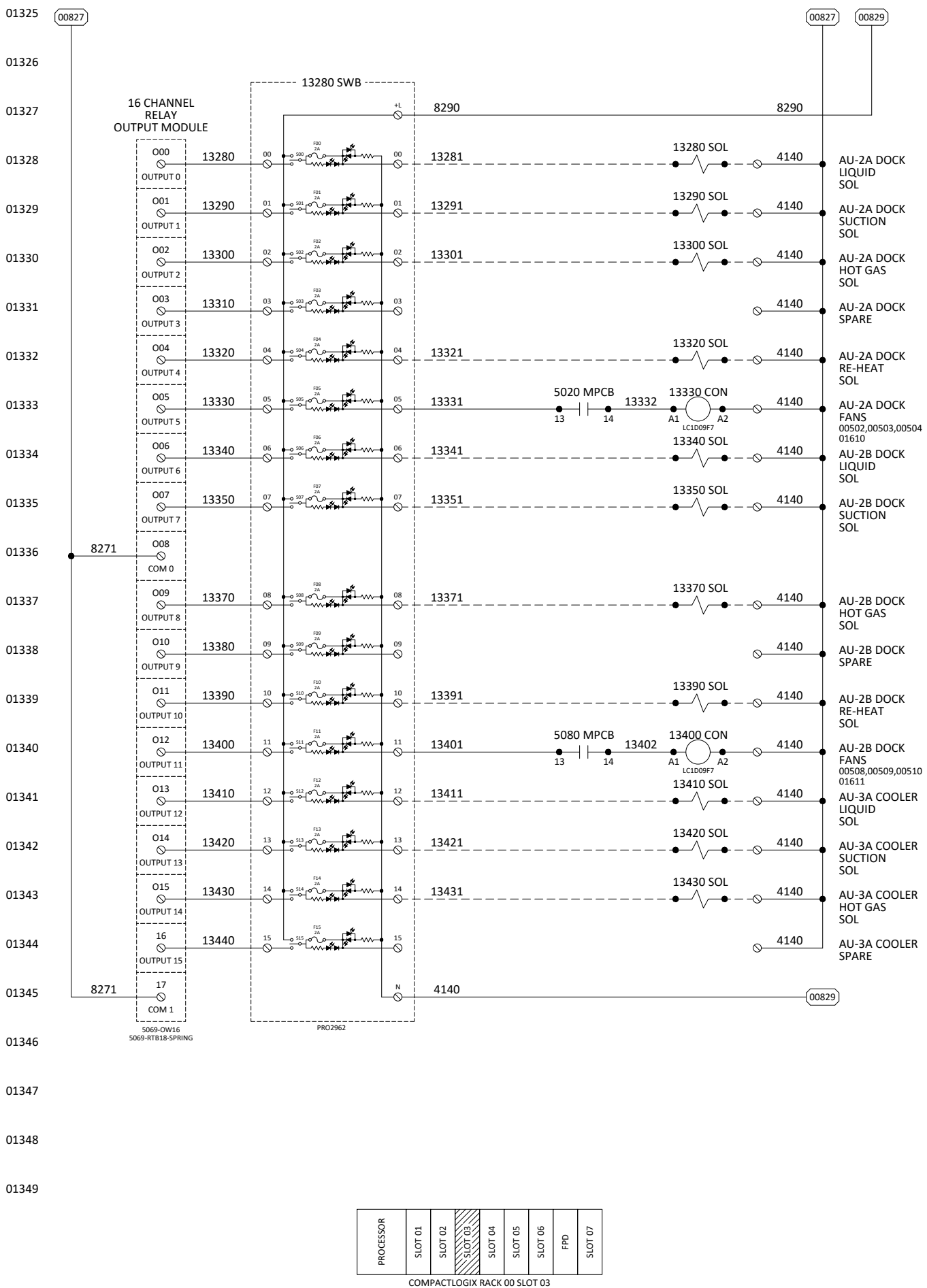
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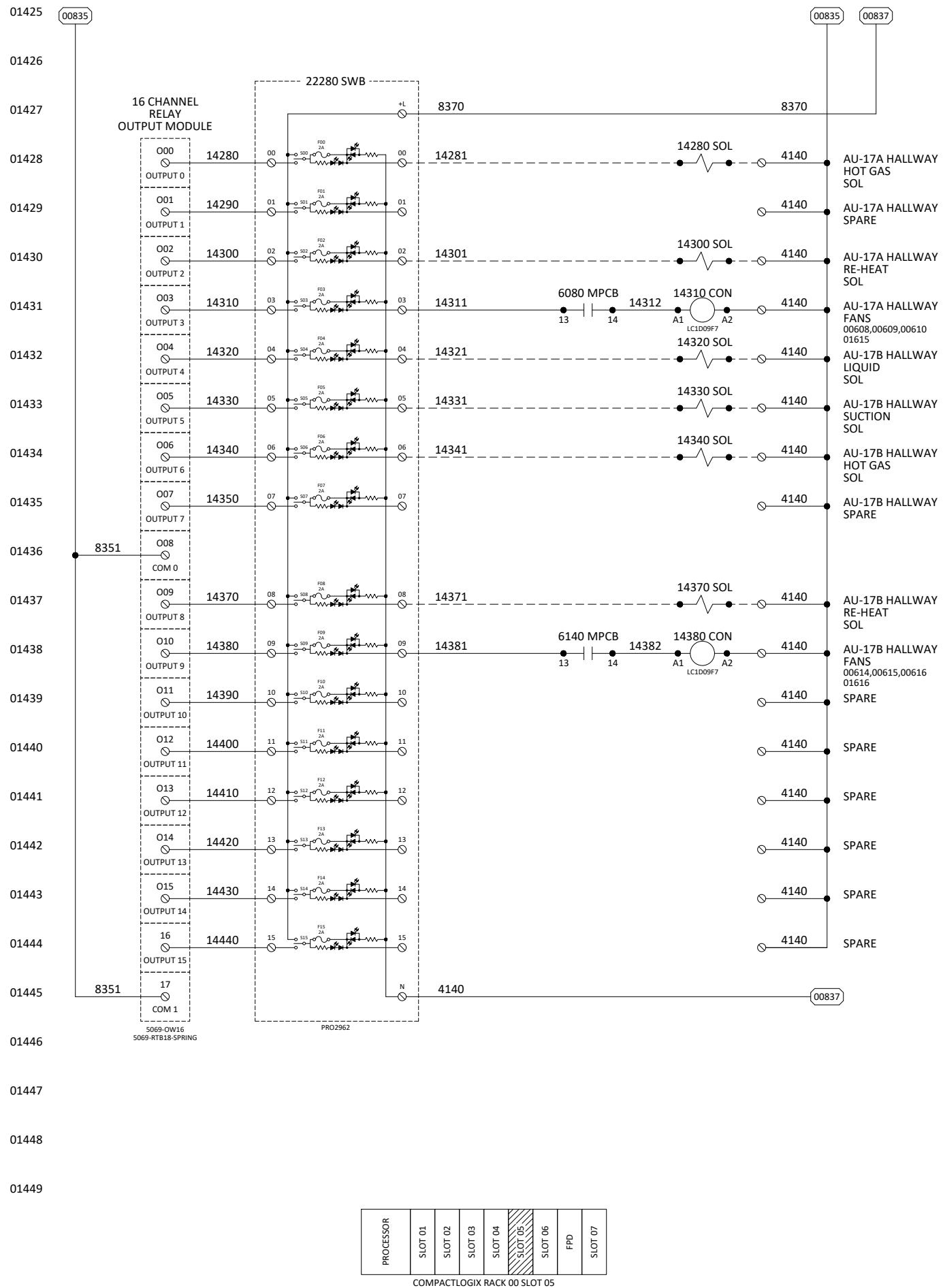
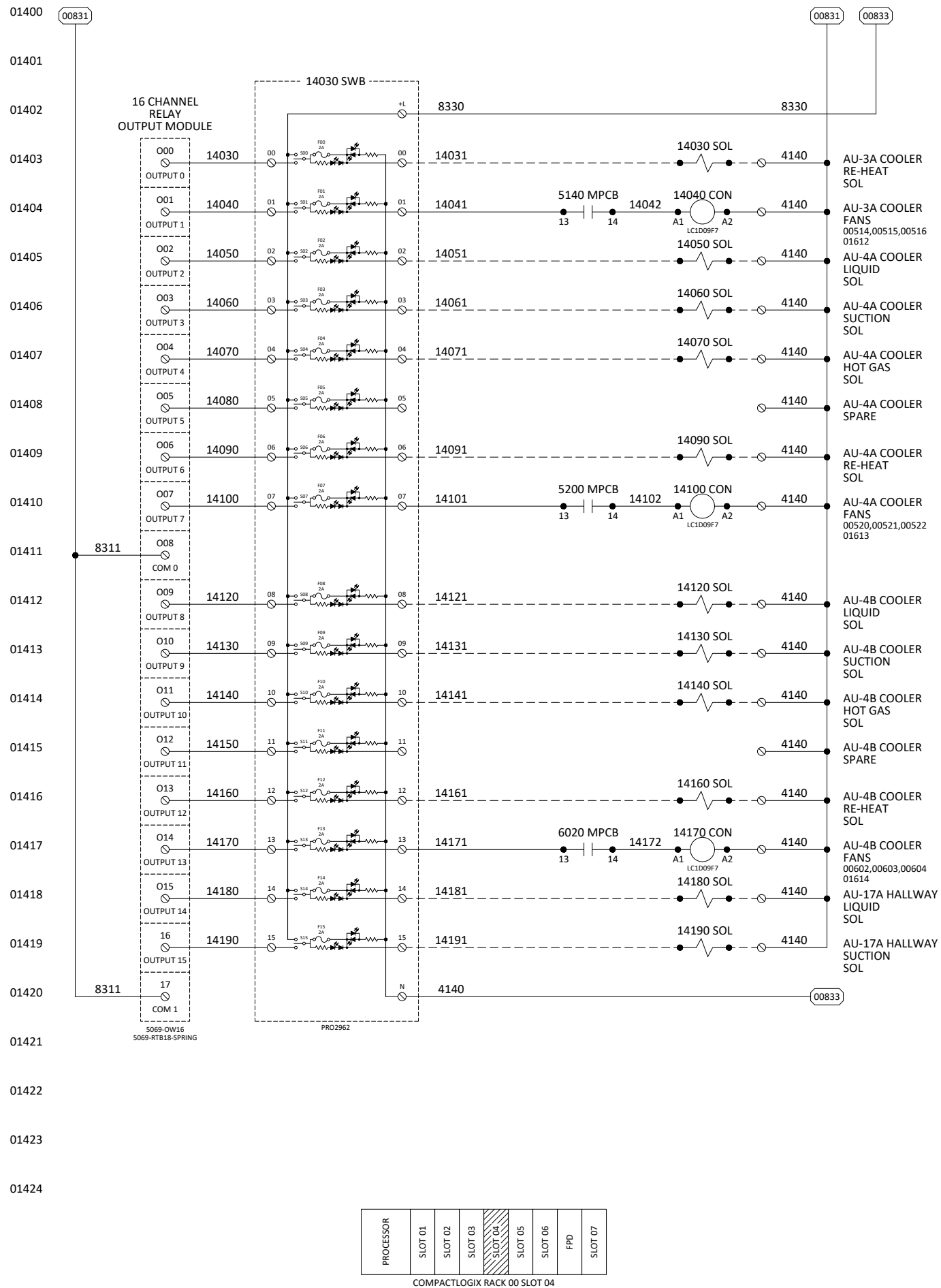


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RACK 00			
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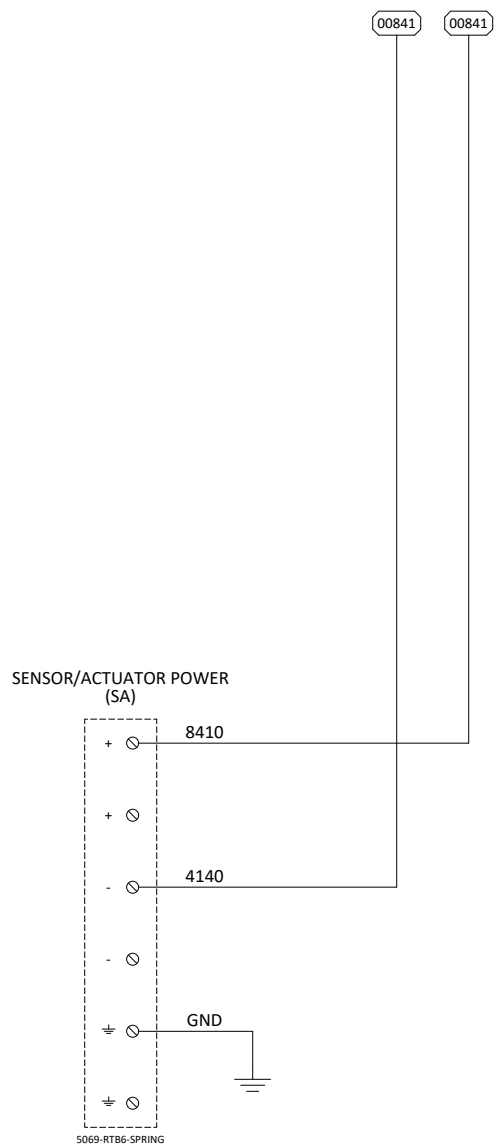
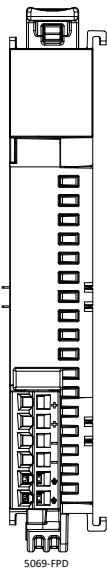
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SLOTS 04 AND 05

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PROCESSOR							
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	SLOT 02						
	SLOT 03						
	SLOT 04						
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	SLOT 06						
						FPD	
							SLOT 07

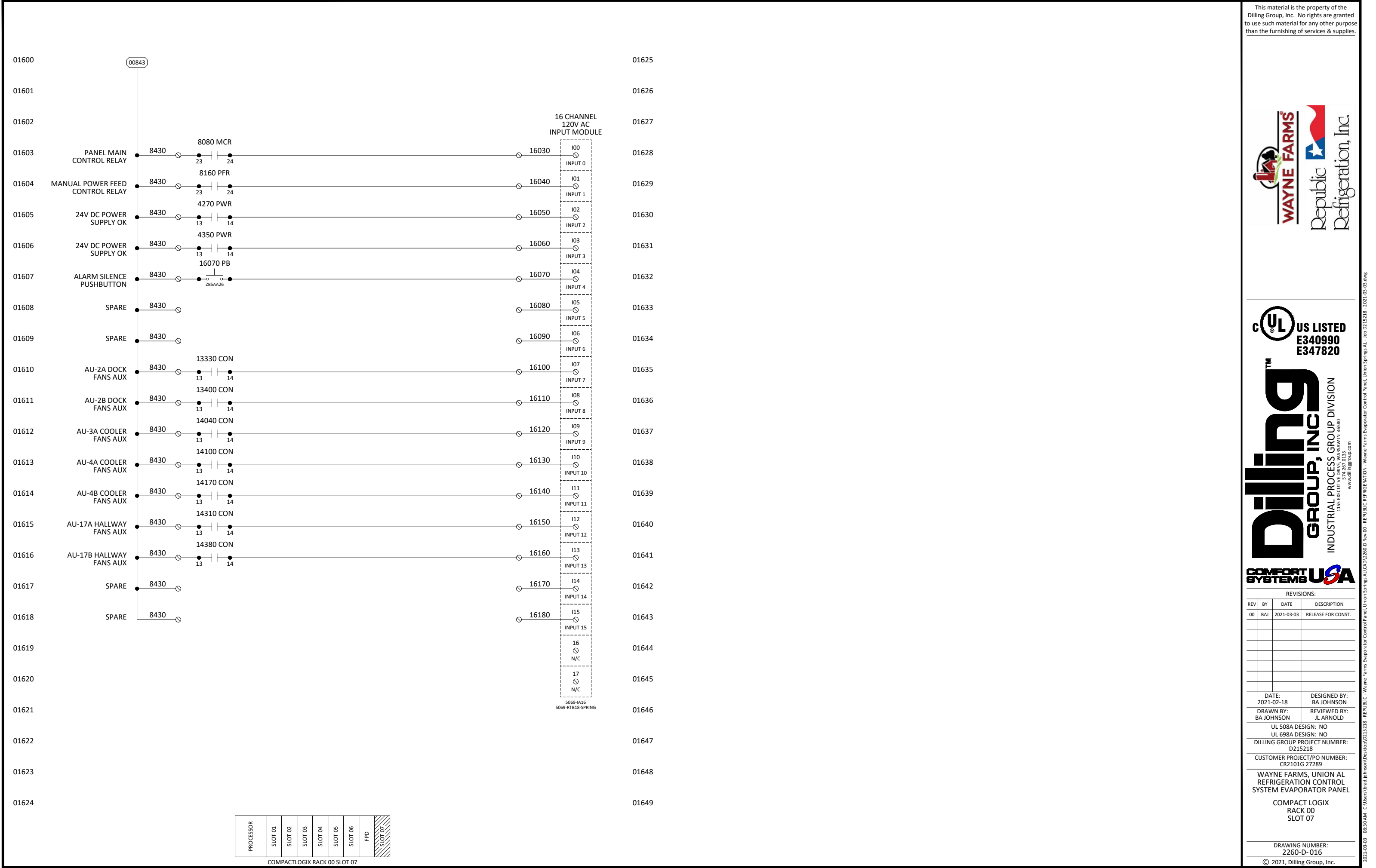
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COMPACTLOGIX RACK 00 SLOT 07

PROCESSOR

SLOT 01

SLOT 02

SLOT 03

SLOT 04

SLOT 05

SLOT 06

FPD

SLOT 07

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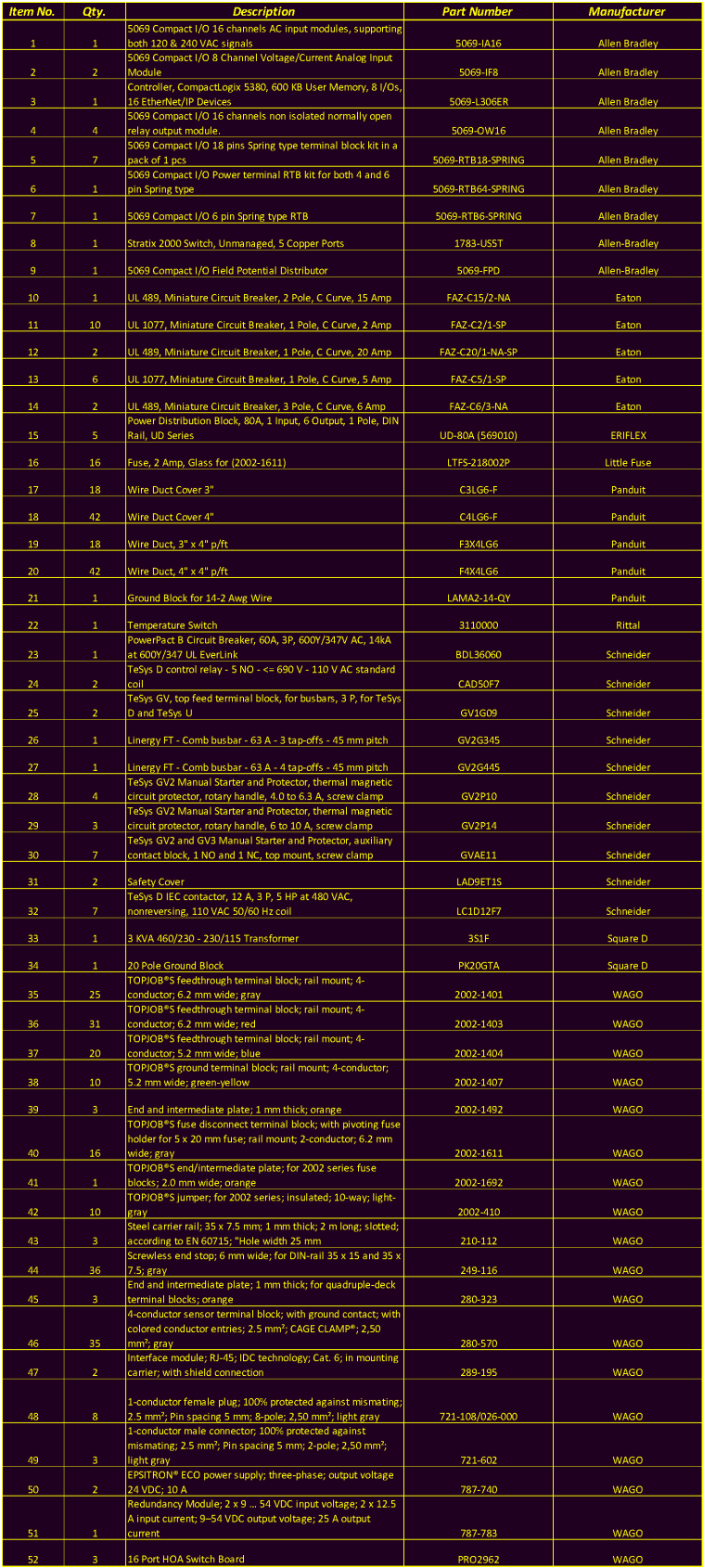
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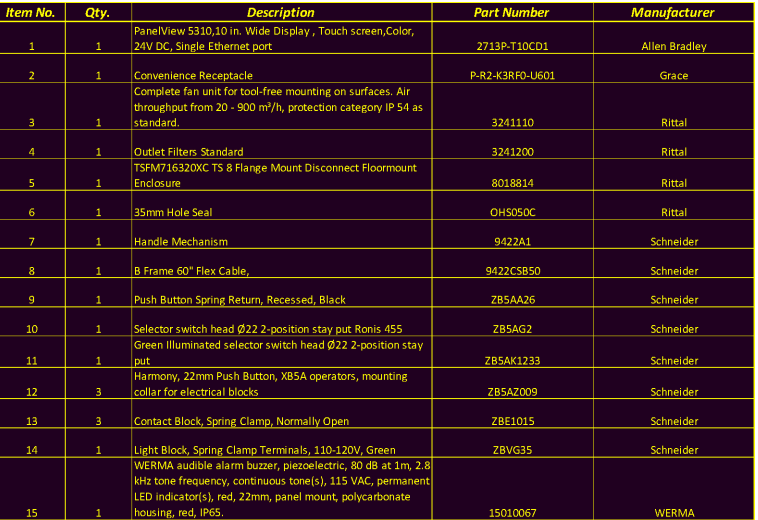
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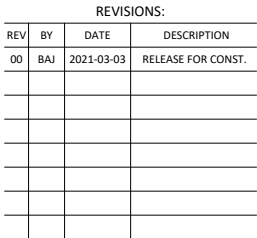
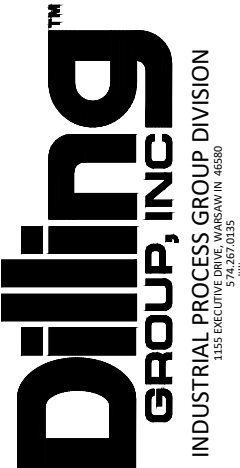


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WAYNE FARMS



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